

Jiuru Lyu

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EDUCATION

B.S. in Applied Mathematics and Statistics, Minor in Computer Informatics

Aug 2022 – May 2026

Emory University, Emory College of Arts and Sciences

Atlanta, GA

- **Cumulative GPA:** 3.99/4.0, **Major GPA:** 4.0/4.0; Dean's List
- **Coursework:** Iterative Methods (Grad), Numerical Analysis (Grad), Numerical PDEs (Grad), Measure Theory (Grad), Real Analysis, Optimization Methods, Mathematical Statistics, Differential Equations, Quantum Computing, Machine Learning, Deep Learning, Regression and Causal Inference, Data Structures & Algorithms, Database System
- **Research Interests:** Numerical Optimization, Numerical Linear Algebra, Machine Learning, Numerical PDEs

SKILLS AND INTERESTS

- **Skills:** $\text{\LaTeX}$, MATLAB, Python, R, Java, PostgreSQL, HTML, CSS, JavaScript, TypeScript, Angular
- **Languages:** Mandarin Chinese (Fluent), English (Fluent), Spanish (Elementary)
- **Interests:** Specialty Coffee, Traveling, Photography, Hiking, Musicals, Mathematics Through Movement

RESEARCH EXPERIENCES

Research Assistant: Global Optimization

Oct 2023 – Present

Department of Mathematics, Emory University

Atlanta, GA

- Conducted independent research on Double Descent with Color Intermittent Diffusion (DD-CID) and other Global Optimization methods under the supervision of Dr. Manuela Manetta; to develop results into an honors thesis
- Implemented a log-barrier regularization and the continuation method to enhance the algorithm for finding Morse Potential problems with high-dimensionality; cross-validated outcomes against the Cambridge Database
- Applied the algorithm to COVID-19 modeling; recasted ODE constraints to an adjoint-based dual form for efficient optimization; improved the ODE solver by manually employing Crank-Nicolson method
- Utilized the land exploring nature of the algorithm to solving benchmark PINN problems, aiming to overcome the bottleneck of getting stuck at saddle points and inability to find global optimum solutions
- Explored ways to integrate derivative-free optimization techniques into the algorithm to enhance efficiency

Research Assistant: PDHG Algorithm for Non-Potential Mean-Field Games

Feb 2025 – Oct 2025

Department of Mathematics, Emory University

Atlanta, GA

- Conducted research on a PDHG solver for non-potential mean-field games supervised by Dr. Levon Nurbekyan
- Developed a general-purpose Python implementation of Primal-Dual Hybrid Gradient (PDHG) solver for non-potential Mean-Field Games (MFGs) with space-time discretization using sparse linear algebra package SciPy
- Assembled discrete operators and block matrices; engineered convergence diagnostics and invariants for MFG structure; accelerated the solver via iterative methods and matrix caching; parallelized the solver

Research Assistant: Spectral Graph Theory

Jan 2025 – Jul 2025

Department of Mathematics, Emory University

Atlanta, GA

- Studied the relationship between Spectral Graph Theory and ODEs under the guidance of Dr. Manuela Manetta
- Led weekly discussion sessions on papers related to spectral graph theory and their relationships with ODEs
- Implemented centrality and other Network Analysis pipelines for the Emory Mathematics Department faculty network; identified communities/bridges and summarized collaboration insights
- Compared symmetrization schemes for directed graphs and their impact on diffusion and ODE dynamics

AWARDS AND HONORS

Trevor Evans Award <i>Department of Mathematics, Emory University</i>	May 2025 Atlanta, GA
Phi Beta Kappa Honor Society <i>Emory Chapter of Phi Beta Kappa (Gamma of Georgia)</i>	Apr 2026 Atlanta, GA
Wilson Family Undergraduate Research Award <i>Department of Mathematics, Emory University</i>	May 2025 Atlanta, GA
Emory Pathway Internship Funding <i>Pathway Center, Emory University</i>	Jun 2024 Atlanta, GA
DataFest Competition Finalist <i>Pathway Center, Emory University</i>	Apr 2024 Atlanta, GA
Phi Eta Sigma Honor Society <i>Emory Chapter of Phi Eta Sigma National Honor Society</i>	Feb 2023 Atlanta, GA

TEACHING/MENTORING EXPERIENCES

Emory Department of Mathematics <i>Grader in MATH 212 (Differential Equations)</i>	Jan 2026 – Present Atlanta, GA
Academic Success, Emory Office of Undergraduate Education <i>Peer Tutor in Mathematics</i>	Sep 2023 – Present Atlanta, GA
Emory Department of Data and Decision Sciences <i>Teaching Assistant for DATASCI 151 (Scientific Computing with Python)</i>	Aug 2024 – Dec 2024 Atlanta, GA

INTERNSHIPS

Baker Hughes <i>Digital Technology Intern</i>	Jun 2024 – Aug 2024 Shanghai, China
- Applied Agile development workflows to a valve lifecycle management system (VKAM); implemented soft-delete and advanced-filter features; improved customer report UI designs for better user experience - Wrote and executed 100+ regression test cases, gained hands-on experience in database validation; discovered 30+ bugs or outdated tests; helped maintain consistent test case writing styles	
Algory Capital <i>Quantitative Investment Analyst</i>	Jan 2023 – Aug 2024 Atlanta, GA
- Oversaw risk assessment of Algory Capital, a student-managed fund with \$100,000 AUM; implemented Monte Carlo Simulations to calculate Conditional Value at Risk (cVaR); integrated insights into a grid search algorithm - Developed advanced trading strategies (e.g., mean reversion pair trading and ADF tests); performed sentiment analysis using Reddit API and web-scraping tools to inform portfolio management decisions	

MANUSCRIPTS AND PUBLICATIONS

Theses	
[T1]	Lyu, Jiuru. <i>Double Descent and Color Intermittent Diffusion: A Global Optimization Framework Across Disciplines</i> . Bachelor’s Honors Thesis, Emory University, 2026. Emory Theses and Dissertations.

CONFERENCE PRESENTATIONS

[C1]	2026 Georgia Scientific Computing Symposium <i>Test Double Descent with Color Intermittent Diffusion (DD-CID) with Challenges in Different Disciplines</i>	Feb 21, 2026 Athens, GA
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